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Question Paper Code : 15743

B.E. / B.TECH. DEGREE EXAMINATION, NOVEMBER / DECEMBER 2019

Third Semester

B.E. – Computer Science and Engineering

15CS303 – COMPUTER ORGANIZATION AND ARCHITECTURE

(Regulations: Mepco – R2015)

Duration: 3 Hours Max. : 100 Marks

Answer ALL Questions

BTL, CO

PART A – (10 × 2 = 20 Marks)

U,CO1 1. The instruction R_2<- [Loc] in register transfer notation is equivalent to _____. Justify.

A) Add R2, R2, LOC B) Add LOC, R2, LOC

C) Store R2, LOC D) Load R2, LOC

R,CO1 2. State the rules to add two integers.

U,CO3 3. The register which holds the return address during a subroutine call is _____. Justify.

A) LINK B) PC C) PC-Temp D) RA

A,CO4 4. Consider a RISC machine where each instruction is exactly 4 bytes long. Conditional and unconditional branch instructions use PC-relative addressing mode with offset specified in bytes to the target location of the branch instruction. Further the offset is always with respect to the address of the next instruction in the program sequence. Consider the following instruction sequence:

Instruction No.	Instruction
i:	ADD R2,R3,R4
i+1:	SUBTRACT R5,R6,R7
i+2:	BGT R1,R9,TARGET

If the target of the branch instruction is i, then the decimal value of the offset is _____. Justify.

U,CO4 5. What is meant by branch prediction?

R,CO3 6. Write the logic expression for the control signal RF-Write.

A,CO5 7. Consider a two–level cache hierarchy with L1 and L2 caches. An application incurs 1.4 memory access per instruction on average. For this application, the miss rate of L1 cache is 0.1; the L2 cache experiences on average 7 misses per 1000 instructions. The miss rate of L2 expressed correct to two decimal places is _____. Justify.

U,CO5 8. What is the use of DMA Controller?

A,CO6 9. A processor can support a maximum memory of 4GB, where the memory is word-addressable (a word consists of two bytes). The size of address bus of the processor is atleast _____ bits. Justify.

U,CO6 10. What is meant by memory-mapped I/O?

PART B – (5 × 16 = 80 Marks)

U,CO1 11. a) i. With a neat diagram, explain how looping is implemented using Branch instruction. (8 Marks)

A,CO2 11. a) ii. Add the numbers $(0.65)_{10}$ and $(-0.456)_{10}$ in binary using the floating point addition algorithm. (8 Marks)

OR

U,CO1 11. b) i. What is an addressing mode in a computer? Classify RISC addressing modes and give one example instruction to each category. (8 Marks)

A,CO2 11. b) ii. Using Booth's algorithm perform multiplication between

A) 38 and 24 B) 27 and -19 (8 Marks)

U,CO3 12. a) i. At the time the instruction Load R_6, 1000(R_9) is fetched, R_6 and R_9 contain the values 4200 and 85320 respectively. Memory location 86320 contains 75900. Show the contents of the interstage registers in datapath of a RISC style processor during each of the 5 execution steps of this instruction. (8 Marks)

U,CO3 12. a) ii. Design a simple datapath of a RISC style processor and explain it in detail. (8 Marks)

OR

A,CO3 12. b) i. At some point in the execution of a program, registers R4, R6 and R7 contain the values 1000, 7500 and 2500 respectively. Show the contents of the interstage registers in datapath of a RISC style processor during steps 1 to 5 as the instruction SUBTRACT R6, R4, R7 is fetched and executed. (8 Marks)

U,CO3 12. b) ii. Explain how hardwired control is used to generate control signals. Also state and explain the equations of any two control signals. (8 Marks)

A,CO4 13. a) i. The following sequence of instructions are executed in the basic 5-stage pipelined processor:

Load R1, 40(R6)

ADD R6,R2,R2

STORE R6,LOC

A) Indicate dependencies and their type

B) Assume there is no forwarding in this pipelined processor. Indicate hazards and add NOP instructions to eliminate them

C) Assume there is full forwarding. Indicate hazards and add NOP instructions to eliminate them (8 Marks)

U,CO4 13. a) ii. What are the various situations where an instruction pipeline can stall? Illustrate with an example. (8 Marks)

OR

A,CO4 13. b) i. Assume that 30% of the dynamic count of the instructions executed for a program are branch instructions. There are no pipleline stalls due to data dependencies. Static branch prediction is used with a not-taken assumption.

A) Determine the execution time for the two cases: when 45% of the branches are taken and when 80% of the branches are taken.

B) Determine the speed up for one case relative to the other. Express the speed up as a percentage relative to 1.03. (8 Marks)

U,CO4 13. b) ii. Explain in detail the pipeline performance evaluation. (8 Marks)

A,CO5 14. a) i. A block-set-associative cache consist of a total of 128 blocks, divided into 8 block sets. The main memory contains 16384 blocks, each consisting of 256 words.

A) How many bits are there in the main memory address?

B) How many bits are there in each of the TAG, SET and WORD fields? (8 Marks)

U,CO5 14. a) ii. Explain Dynamic RAMs in detail. (8 Marks)

OR

A,CO5 14. b) i. Assume a computer has L1 and L2 caches, the cache blocks consist of 8 words. Assume that the hit rate is the same for both caches and that it is equal to 0.95 for instructions and 0.90 for data. Assume also that the time needed to access an 8-word block in these caches are C1=1 cycles and C2=10 cycles.

A) What is the average access time experienced by the processor, if the main memory uses interleaving?

B) What is the average access time, if the main memory is not interleaving? (8 Marks)

U,CO5 14. b) ii. Explain in detail about Direct Memory Access. (8 Marks)

U,CO5 15. a) i. Explain about parallel interface circuits for input device and its functions in detail. (8 Marks)

U,CO6 15. a) ii. Describe in detail about standard I/O interface PCI. (8 Marks)

OR

U,CO6 15. b) i. Explain the handshake control of data transfer during an input operation using Asynchronous Bus with a neat diagram. (8 Marks)

U,CO6 15. b) ii. Explain in detail about program controlled I/O. (8 Marks)